
```
clc
clear
close all
```

Problem 1

```
%b)
clc
clear
a=.3;
b=.1;
delta_0=[1,0,0,0,0];
p_1=zeros(5,5);
p_1(1,1)=1-a;
p_1(1,2)=a;
for i=2:4

    p_1(i,i+1)=a.*(4-i)./i;
    p_1(i,i-1)=b.*i./4;
    p_1(i,i)=1-p_1(i,i+1)-p_1(i,i-1);

end
p_1
q_50=delta_0*(p_1^50)
q_51=delta_0*(p_1^51)
n=1:60;
for i=1:60

    q=delta_0*(p_1^i);
    q_2(i)=q(2);
    q_3(i)=q(3);
    q_ave_2(i)=mean(q_2);
end
[V,lambda]=eig(p_1)
figure
hold on
plot(n,q_2)
plot(n,q_3)
plot(n,q_ave_2)
legend('q_2','q_3','running average')
xlabel('n')
ylabel('q value')
hold off
for i=0:4
    pi(i+1)=((.25).^(4-i)).*((3./4).^i).*factorial(4)./(
        factorial(i).*factorial(4-i))
end
```

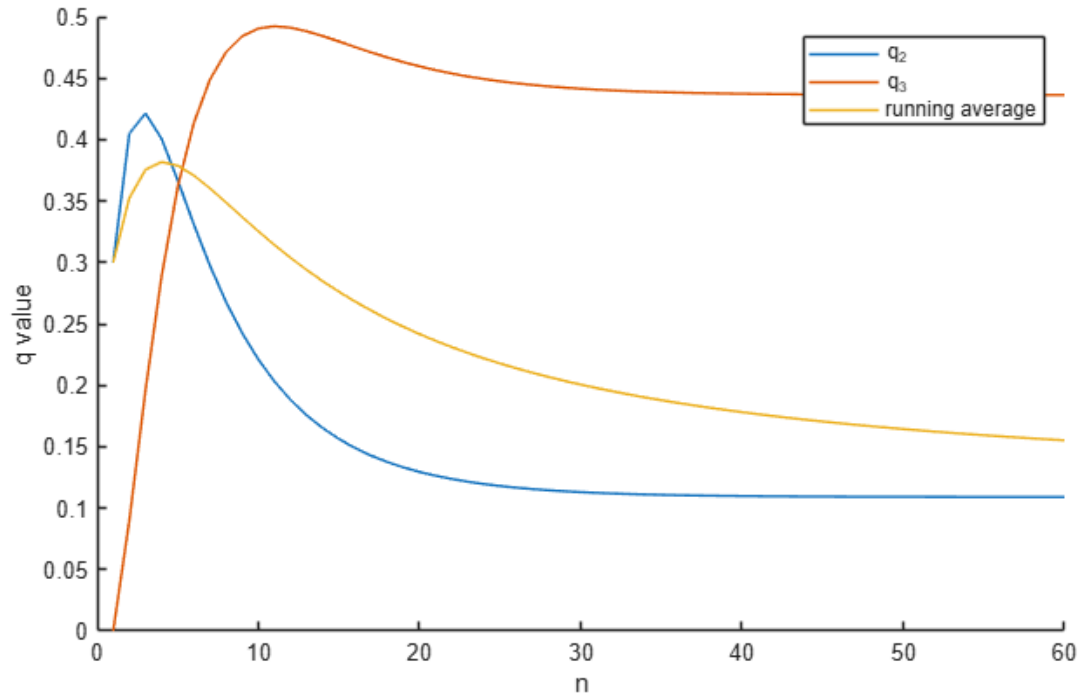
```
p_1 =

    0.7000    0.3000         0         0         0
```

0.0500	0.6500	0.3000	0	0
0	0.0750	0.8250	0.1000	0
0	0	0.1000	0.9000	0
0	0	0	0	0
$q_{50} =$				
0.0182	0.1092	0.4366	0.4360	0
$q_{51} =$				
0.0182	0.1092	0.4365	0.4360	0
$V =$				
0.8237	0.9829	-0.8485	0.5000	0
-0.5491	0.0819	-0.4243	0.5000	0
0.1373	-0.1433	-0.1414	0.5000	0
-0.0343	0.0819	0.2828	0.5000	0
0	0	0	0	1.0000
$\lambda =$				
0.5000	0	0	0	0
0	0.7250	0	0	0
0	0	0.8500	0	0
0	0	0	1.0000	0
0	0	0	0	0
$\pi =$				
0.0039				
$\pi =$				
0.0039	0.0469			
$\pi =$				
0.0039	0.0469	0.2109		
$\pi =$				
0.0039	0.0469	0.2109	0.4219	

$pi =$

0.0039 0.0469 0.2109 0.4219 0.3164



Problem 3

a)

```
p_a=mat_p(1); %Checking
p_test=mat_p(.6); %Testing
q_0=[1,1,1,1,1,1,1,1]./8;
q_100=q_0*(p_a^100)
q_101=q_0*(p_a^101)
%does not converge
%{
q_100_test=q_0*(p_test^100)
q_101_test=q_0*(p_test^101)
%converges
%}

%b
d=.85

p=mat_p(d)
diff=1;
n=1;
q_n1=q_0;
while diff>10e-6
```

```
q_n=q_n1;
q_n1=q_n*p;
diff=mean(abs(q_n1-q_n))
n=n+1;
if n>1000
    break
end
```

```
end
q_n
q_n1
n
```

```
% c)
```

```
%{
function [y,t]=discrete(U,t,p,i)
V=rand(1);
t=t+1;
steps(1)=p(i,1);
for j=1:8
steps(j+1)=steps(j)+p(i,j);
end
if U<=steps(1)
    [y,t]=discrete(V,t,p,i);
elseif U<=steps(2)
    [y,t]=discrete(V,t,p,i);
elseif U<=steps(3)
    [y,t]=discrete(V,t,p,i);
elseif U<=steps(4)
    [y,t]=discrete(V,t,p,i);
elseif U<=steps(5)
    [y,t]=discrete(V,t,p,i);
elseif U<=steps(6)
    [y,t]=discrete(V,t,p,i);
elseif U<=steps(7)
    [y,t]=discrete(V,t,p,i);
else
    [y,t]=discrete(V,t,p,i);
end

end
```

```
%}
```

```
function p=mat_p(d)
p=[0,d./2,d./2,0,0,0,0,0;0,0,0,d,0,0,0,0;d./2,0,0,0,d./
```

```

2,0,0,0;d./3,0,d./3,0,d./3,0,0,0;0,d./3,0,0,0,d./3,d./
3,0;0,0,0,0,0,0,d,0,0;0,0,0,0,0,0,0,d;0,0,0,0,0,0,d,0];
p=p+(1-d)./8;

```

```

end
%{
function steps=discete(P,t)
[m,n]=size(P);
steps=zeros(n,1);
for k=1:t
for i=1:m
s=zeros(n,1);
V=rand(1);
j=1;
while j<n+1

    if V>s(j,1)
s(j,1)=P(i,j)+s(j,1);
j=j+1;
else
steps(j,1)=steps(j,1)+1;
break
end
end
end
end
end

%}

```

```

function x=inverse_transform(P,X)

```

```

end

```

```

q_100 =

```

```

Columns 1 through 7

```

```

0.0000    0.0000    0.0000    0.0000    0.0000    0.4375    0.2695

```

```

Column 8

```

```

0.2930

```

```

q_101 =

```

```

Columns 1 through 7

```

```

0.0000    0.0000    0.0000    0.0000    0.0000    0.4375    0.2930

```

Column 8

0.2695

$d =$

0.8500

$p =$

Columns 1 through 7

0.0188	0.4437	0.4437	0.0188	0.0188	0.0188	0.0188
0.0188	0.0188	0.0188	0.8688	0.0188	0.0188	0.0188
0.4437	0.0188	0.0188	0.0188	0.4437	0.0188	0.0188
0.3021	0.0188	0.3021	0.0188	0.3021	0.0188	0.0188
0.0188	0.3021	0.0188	0.0188	0.0188	0.3021	0.3021
0.0188	0.0188	0.0188	0.0188	0.0188	0.8688	0.0188
0.0188	0.0188	0.0188	0.0188	0.0188	0.0188	0.0188
0.0188	0.0188	0.0188	0.0188	0.0188	0.0188	0.8688

Column 8

0.0188
0.0188
0.0188
0.0188
0.0188
0.0188
0.8688
0.0188

$diff =$

0.0177

$diff =$

0.0138

$diff =$

0.0107

$diff =$

0.0085

diff =

0.0064

diff =

0.0051

diff =

0.0040

diff =

0.0032

diff =

0.0025

diff =

0.0020

diff =

0.0016

diff =

0.0013

diff =

0.0010

diff =

8.6140e-04

diff =

6.9436e-04

diff =

5.7627e-04

diff =

4.6925e-04

diff =

3.9127e-04

diff =

3.2136e-04

diff =

2.6902e-04

diff =

2.2256e-04

diff =

1.8692e-04

diff =

1.5555e-04

diff =

1.3099e-04

diff =

1.0953e-04

diff =

9.2431e-05

diff =

7.7579e-05

diff =

6.5577e-05

diff =

5.5202e-05

diff =

4.6723e-05

diff =

3.9422e-05

diff =

3.3400e-05

diff =

2.8230e-05

diff =

2.3937e-05

diff =

2.0259e-05

diff =

1.7188e-05

diff =

1.4563e-05

diff =

1.2361e-05

diff =

1.0481e-05

diff =

8.8991e-06

q_n =

Columns 1 through 7

0.0707	0.0688	0.0707	0.0772	0.0707	0.2585	0.1972
--------	--------	--------	--------	--------	--------	--------

Column 8

0.1863

q_n1 =

Columns 1 through 7

0.0707	0.0688	0.0707	0.0772	0.0707	0.2585	0.1971
--------	--------	--------	--------	--------	--------	--------

Column 8

0.1863

n =

41

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